

# Pengambilan Keputusan Berbasis Probabilitas dan Statistika

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# Daftar Isi

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# Preface

This is a Quarto book.

To learn more about Quarto books visit <https://quarto.org/docs/books>.



# 1

## Introduction

This is a book created from markdown and executable code.

See D. E. Knuth [1] for additional discussion of literate programming.

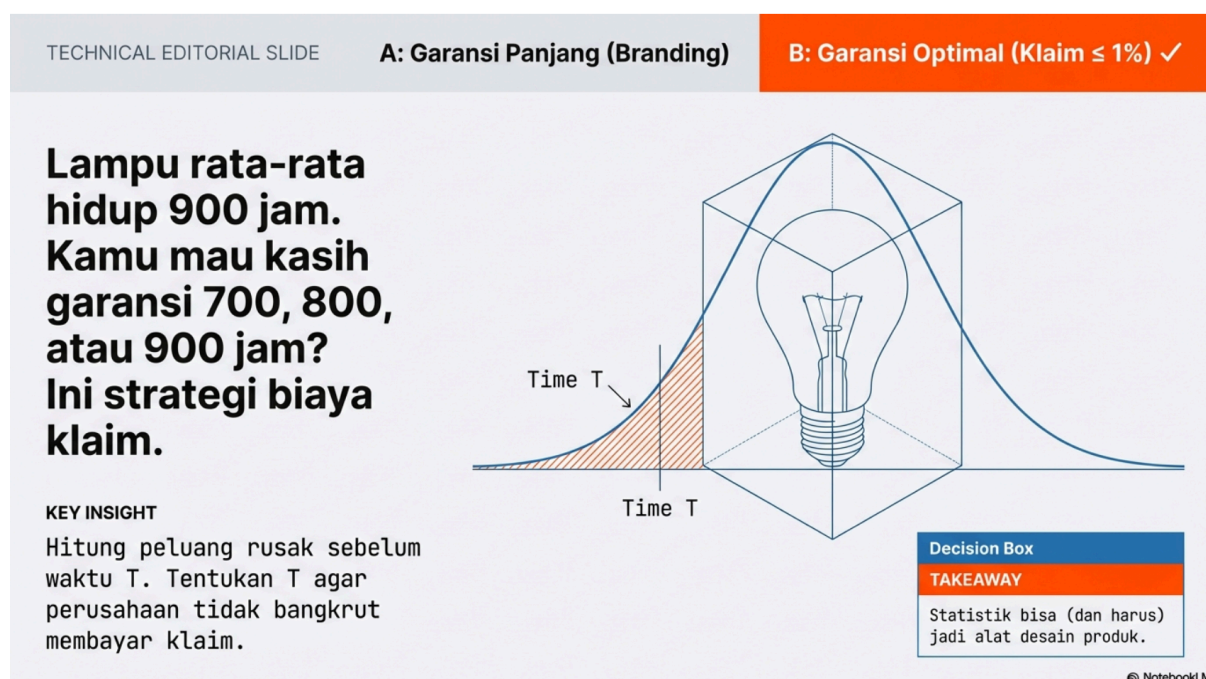




# 2

## Random Variable Kontinu

### 2.1. Kita Beri Garansi Berapa Lama?



Gambar 2.1 – “Lampu rata-rata hidup 900 jam, simpangan 50 jam. Kamu mau kasih garansi 700 jam, 800 jam, atau 900 jam? Ini bukan sekadar ‘baik hati’—ini strategi biaya klaim. Normal sering muncul di data pengukuran; eksponensial sering muncul di waktu tunggu. Hari ini kamu belajar menghitung peluang rusak sebelum waktu T, lalu menentukan T yang masuk akal untuk target klaim. Kamu akan melihat: statistik bisa jadi alat desain produk.”

## 2.2. Live Coding: “Normal is Everywhere”\*\*

Suatu variable random bisa memiliki *probability density function* (pdf) berbentuk lonceng, dengan dua parameter  $\mu$  dan  $\sigma$ .

```
import matplotlib.pyplot as plt
import numpy as np

# Instantiate a default random number generator
rng = np.random.default_rng()
```

Untuk distribusi normal dan standar, lokasi titik tengah = 0 dan deviasi standar = 1.

```
# Generate a single random number from the standard normal distribution (mean=0,
std=1)
sample_scalar = rng.normal()
print(f"1. Single sample: {sample_scalar}")
```

```
1. Single sample: 0.24416642202191122
```

Dalam kasus kita lokasi titik tengah = 900 dan deviasi standar = 50. maka berapa jam lima lampu ini mati?

```
mu = 900
sigma = 50

# Generate a 1D array of 5 numbers with mean=900, standard deviation=50
samples_1d = rng.normal(loc=mu, scale=sigma, size=5)
print(f"2. 1D array: {samples_1d}")
```

```
2. 1D array: [ 921.64135141  981.44396981  858.85745202 1035.86023778  972.51206834]
```

Berbeda beda yaitu: `np.float64(921.6413514058673)`

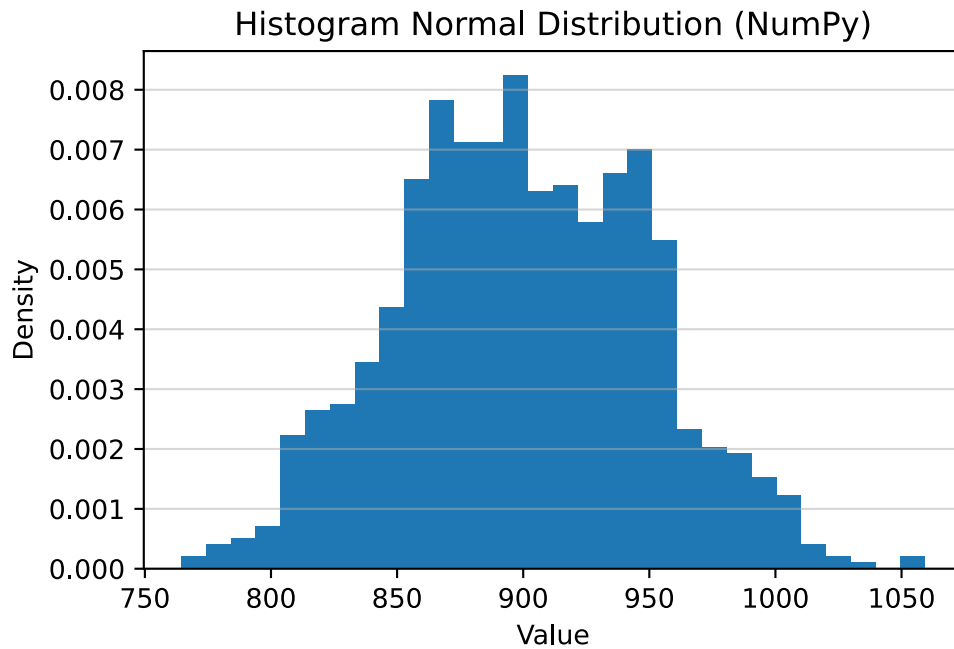
1. Generate data acak ``numpy.random.normal``.
2. Tunjukkan aturan empiris: 68% data ada di  $\mu \pm \sigma$ , 95% di  $\mu \pm 2\sigma$ .
3. Visualisasi: Plot kurva lonceng dengan seaborn dan arsir area probabilitas.

```
data = rng.normal(loc=mu, scale=sigma, size=1000)
np.histogram(data, bins=30, density=True)

# 3. Menambahkan label dan judul
plt.hist(data, bins=30, density=True)
plt.title('Histogram Normal Distribution (NumPy)')
plt.xlabel('Value')
plt.ylabel('Density')
```

```
plt.grid(axis='y', alpha=0.5)
```

```
# 4. Menampilkan plot  
plt.show()
```



```
# Generate a 2x3 matrix with default parameters  
samples_2d = rng.normal(size=(2, 3))  
print(f"3. 2D array:\n{samples_2d}")
```

```
3. 2D array:  
[[ 0.65561043  0.12842887  0.17154068]  
 [ 1.22288256 -0.37581476  2.31328241]]
```



# 3

## Summary

In summary, this book has no content whatsoever.



## References





# Daftar Pustaka

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- [1] D. E. Knuth, “Literate Programming,” *Comput. J.*, vol. 27, no. 2, pp. 97–111, May 1984, doi: 10.1093/comjnl/27.2.97.